Prischa Bajeli

AI/ML Engineer | Applied AI & Backend Development

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SUMMARY

AI & Data Science undergraduate (2027) with internship experience building Python/FastAPI services, multilingual speech systems, and ML-driven traffic simulations. Builds RAG assistants, tool-using agents, and computer vision models; first author of two peer-reviewed publications (IEEE, Springer).

EDUCATION

B.Tech, Computer Science Engineering - Artificial Intelligence and Data Science *2023 – 2027 | Pune, Maharashtra*
MIT World Peace University | CGPA: 8.23

Indian School Muscat, Oman (CBSE) | Class XII: 84% | Class X: 94% *2023 | Muscat, Oman*

EXPERIENCE

Choice TechLabs, AI/Web Development Intern *Jul 2026 – Present | Baner, Pune*

- **Worklens:** Built an async FastAPI + PostgreSQL telemetry platform with a cross-platform Python desktop agent, JWT device auth, S3 screenshot pipeline, admin analytics dashboard, and Gemini-powered productivity summaries.
- Designed range-partitioned time-series tables with **20+ Alembic migrations**, and an offline-tolerant agent queue with retry/backfill and signed auto-updates; maintained **38 pytest suites** with mocked AWS (moto) and Dockerized deployment.
- **Vaani:** Built a FastAPI voice-cloning microservice routing **English and 11 Indic languages** to separate TTS engines over HTTP to isolate dependency conflicts, with Postgres-backed voice profiles.
- **Call Quality Analyzer:** Built an **11-stage pipeline** (ASR, sentiment, PII redaction, compliance checks) with a FastAPI backend and an upload-and-analyze web dashboard.

PROJECTS

AI-Powered Emergency Corridor - Python | SUMO/TraCI | FastAPI | PostGIS | LangGraph

- Built a green-corridor system for Pune 108 ambulances on a SUMO + TraCI simulation of central Pune (OpenStreetMap), with traffic-signal pre-emption, live TomTom traffic, Twilio police alerts for jams, and hospital readiness checks.
- Trained scikit-learn ETA and deadlock models, reducing **ETA error from 71.5 s to 7.8 s** in simulation; real-time FastAPI + WebSocket backend on PostGIS with control-room, crew, and admin dashboards and a Claude/LangGraph supervisor agent.

Explainable AI for Chest X-Ray Diagnosis - Python | PyTorch | DenseNet121 | FastAPI | Grad-CAM github link

- Trained a multi-label DenseNet121 on NIH ChestX-ray14 (**112K images, 14 labels**) with patient-level splits to prevent leakage, reaching **0.83 macro-AUC**; tuned per-label thresholds via ROC curves.
- Served the model through a FastAPI REST endpoint with a Grad-CAM web UI showing per-label probabilities and heatmaps.

Siora2k5 – Agentic AI Research Assistant - Python | LangGraph | FastAPI | Next.js | ChromaDB github link

- Built an agent that selects and chains tools (web search, calculation, document RAG, vision) for multi-step questions, implemented both from scratch and with LangGraph.
- Engineered a ChromaDB RAG pipeline behind a FastAPI backend and Next.js chat UI, with an evaluation harness measuring tool-selection accuracy (**100% on the test set**).

JobAI – AI-Powered Career Copilot - TypeScript | React Native | Express.js | PostgreSQL | Drizzle ORM github link

- Developed a cross-platform mobile career assistant in React Native/Expo, using the Claude API to tailor resumes, cover letters, and outreach messages.
- Built a type-safe Express.js + PostgreSQL + Drizzle ORM backend with a job-application tracker, and TanStack React Query for client-side caching and sync.

RESEARCH AND PUBLICATIONS

IoT-Enabled Face Recognition Attendance System – First author, IEEE ETCOM 2025. paper link | github link | project blog

- Edge-deployed LBPH face recognition on Raspberry Pi 4 with GPIO integration and SQLite logging, achieving **95%+ accuracy across 50+ student profiles** and cutting per-session attendance logging from ~5 minutes to zero manual input.

Multivariate Time-Series Energy Consumption Prediction (RFE-LSTM) – First author, Springer LNNS, ICTCS 2025, Jaipur. paper link | github link | project blog

- RFE-LSTM forecasting on **35K hourly energy samples**; ablation over feature and window counts cut **MAE/RMSE by up to 7%**.

SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, SQL
AI/ML: PyTorch, TensorFlow, scikit-learn, LangGraph, MCP, RAG, ChromaDB, Claude API, Gemini API, OpenCV, NLP
Backend & Databases: FastAPI, Express.js, PostgreSQL, PostGIS, SQLite, Prisma, Drizzle ORM, Alembic, REST APIs, WebSockets, JWT
Frontend & Mobile: React, Next.js, React Native, Expo, TanStack React Query, Leaflet/MapLibre
Cloud & Tools: AWS (S3), Docker, GitHub Actions (CI/CD), pytest, Git, Raspberry Pi, SUMO/TraCI, Twilio
Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, DBMS, Operating Systems
Certifications: AI Fundamentals – IBM SkillsBuild (2024) | Data Visualization & Dashboards, Excel and Cognos (2024) | Intro to LLMs – NPTEL (2026)